

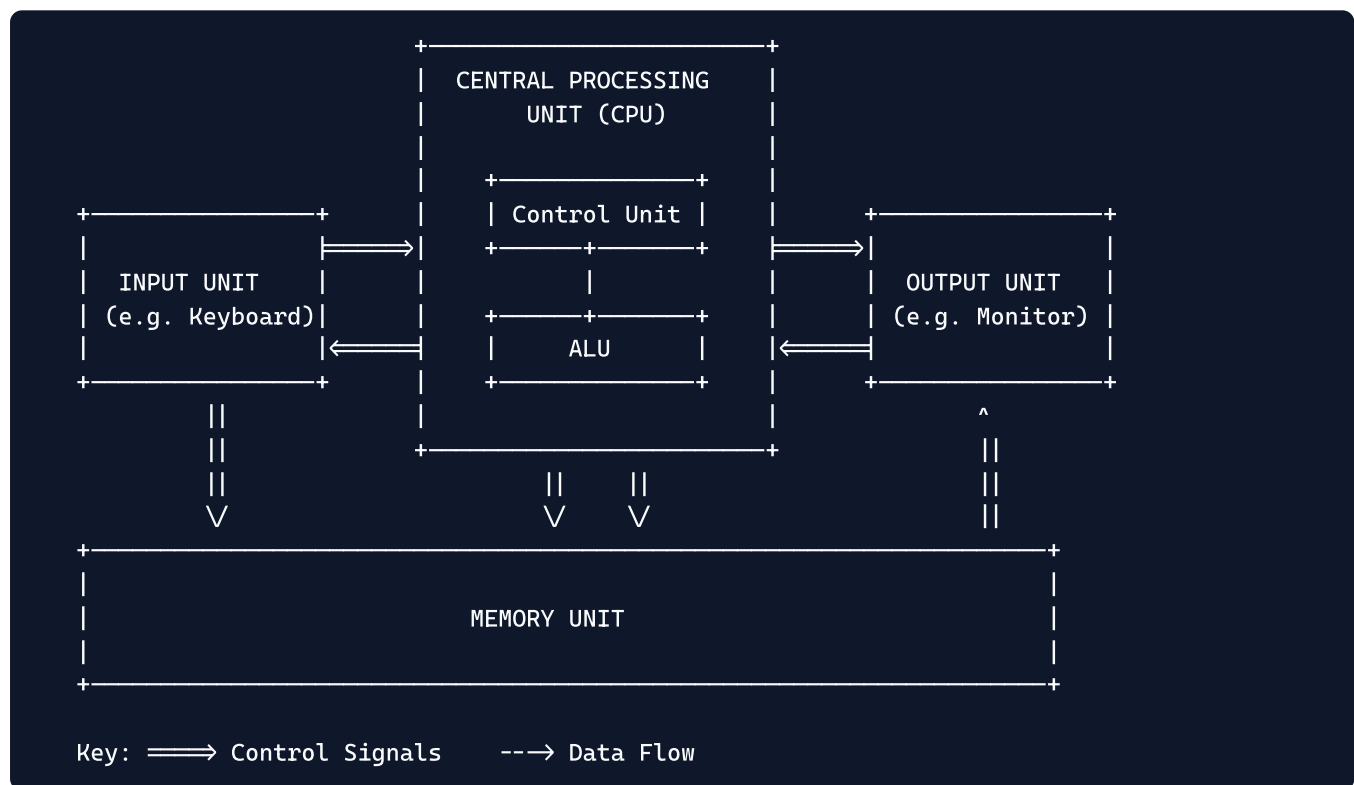
Topic: Block-Level Description of Functional Units — Self-Learning

Q. Draw a block-level diagram of a computer system. Describe how the functional units are interconnected and how data and control signals flow between them.

★ Definition to Remember

A **Block-Level Description** illustrates the physical grouping of a computer's functional units (Input, Output, Memory, CPU) and how they communicate. They do not operate in isolation; they are interconnected via a **Bus System** that transports data, memory addresses, and control signals.

1. Block Diagram of Functional Units



2. The Bus System (Interconnection)

Units communicate over a **Bus System**:

1. **Data Bus:** Bidirectional. Carries actual data between CPU, Memory, and I/O.
2. **Address Bus:** Unidirectional. Carries memory addresses from CPU to Memory/I/O, indicating where to read/write.
3. **Control Bus:** Carries control signals (Read, Write, Interrupt) from the Control Unit to synchronize operations.

3. System Flow

1. **Input:** Control Unit sends a *Read* signal; Input data travels over the *Data Bus* to Memory.

2. **Processing:** Control Unit fetches data via *Data Bus*, ALU processes it, result returns to Memory.
 3. **Output:** Control Unit sends a *Write* signal; Memory data travels via *Data Bus* to the Output Unit.
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★ Must-Write Points (for 10 marks)

1. Functional units are interconnected using a **Bus System**.
 2. **Block Diagram** shows CPU (CU + ALU) connected to Memory, Input, and Output units.
 3. **Data Bus:** Bidirectional; carries actual data.
 4. **Address Bus:** Unidirectional; carries addresses from CPU to Memory/IO.
 5. **Control Bus:** Carries synchronization signals (Read/Write) from the Control Unit.
 6. Control Unit directs traffic via control signals (===>).
 7. Data (--->) always flows through the Memory Unit before going to the CPU or Output.
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⚡ Quick Recall

Block Diagram → CPU, Memory, I/O → Interconnected via Buses → Data Bus (2-way, data) →
Address Bus (1-way, location) → Control Bus (Signals: Read/Write)